



Pedro Ulhoa

Date of birth: 02/02/2000 | **Nationality:** Brazilian | **Phone number:**

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WORK EXPERIENCE

RESEARCH SCHOLAR – HARVARD MEDICAL SCHOOL - SPAULDING REHABILITATION HOSPITAL – 23/09/2025 – 30/04/2026 – BOSTON, UNITED STATES

Under the supervision of **Prof. Paolo Bonato, PhD**, I conducted research at the **Motion Analysis Lab, Spaulding Rehabilitation Hospital — Mass General Brigham**, focusing on the development and improvement of the electronics and control systems of the ExoRoboWalker, a lower-limb exoskeleton designed for gait rehabilitation in individuals with Cerebral Palsy (CP).

- Designed and implemented a control system to investigate motor adaptation during gait in healthy subjects. The control system was consisted of a viscous perturbation controller applied during the swing phase of the gait. The implementation was followed by experimental validation with a control group (12 subjects). Preliminary results were accepted and presented at the IEEE EMBC 2026.
- Implemented the controller algorithm using Simulink.
- During the process, I dealt with MoCap systems to measure gait kinematics and performed calibration of force and EMG sensors.
- I'm still serving as an advisor for the same project.

MASTER'S DEGREE RESEARCH FELLOW – FEDERAL UNIVERSITY OF ESPÍRITO SANTO / CNPQ – 01/08/2023 – 26/06/2026 – VITÓRIA, BRAZIL

Concluded my Master's degree in the Robotics and Biomechanics Laboratory (Final grade: **9.27/10** | **3.7/4**):

- Defended my Master's Dissertation in June: **Control System for Robotic Lower-Limb Prostheses for Transfemoral Amputees Using Surface Electromyography of Antagonist Thigh Muscles**.
- Acted as supervisor of the engineering team, coordinating development efforts across two projects: a **Robotic Leg Prosthesis** and a **Lower-Limb Exoskeleton (ExoRoboWalker v2)**.
- Built a myoelectric signal acquisition system used in both projects.
- Coordinated the development of new electronics for the ExoRoboWalker, a lower-limb exoskeleton for overground gait rehabilitation in children and young adults with Cerebral Palsy (CP), developed at the **Motion Analysis Lab of Spaulding Rehabilitation Hospital/Harvard Medical School**, through an ongoing research collaboration between Harvard University and the Federal University of Espírito Santo on Rehabilitation Robotics.
- Contributed to research publications presented at national and international conferences.
- Conducted validation tests of the robotic leg prosthesis with transfemoral amputees.

CTO - FOUNDER – SYMBIOS – 01/11/2022 – Current – VITÓRIA, BRAZIL

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Link : https://www.instagram.com/reel/DFGr6KOy981/?utm_source=ig_web_copy_link&igsh=MzRIODBiNWFIZA==

- Founded a startup dedicated to advancing the field of Neuromotor Rehabilitation through the development of an Arm Exoskeleton.
- Coordinated the Electronics team during the development of a 3 degree-of-freedom exoskeleton for neurorehabilitation.
- Guided the development of medical technology according to medical standards (IEC60601; ISO13485; IEC80601).
- Planned and executed preliminary clinical tests with patients.
- Secured grant funding in 2023 to initiate the technology and business development.

As an Undergraduate Research Fellow in the Robotics and Biomechanics Laboratory:

- Designed the electrical architecture and control system for a lightweight, backdrivable robotic leg prosthesis for transfemoral amputees.
- Integrated force and myoelectric sensors to enhance prosthesis responsiveness.
- Implemented impedance control algorithms on the embedded processor to improve gait adaptability.
- Contributed to research publications presented at national and international conferences.
- Assisted in managing the electronics team, coordinating development efforts and troubleshooting hardware challenges.

ROBOTICS ENGINEERING INTERN – INWOUND – 01/09/2022 – 15/01/2023 – VILA VELHA, BRAZIL

Robotics Intern at [InWound](#), a Danish startup focused on developing an automated treatment for chronic wounds to help prevent amputations among diabetic patients and other chronic wound patients.

- Developed hardware and firmware for robotics and medical applications.
- Worked with a multidisciplinary and international team of developers.
- Helped a team of engineers build a medical product for certification with regulatory agents.
- Worked on PCB design for medical applications and firmware development (IEC60601)

● EDUCATION AND TRAINING

01/03/2018 – 01/07/2023 Vitória, Brazil

BACHELOR IN ELECTRICAL ENGINEERING Federal University of Espírito Santo

Website <https://www.ufes.br/> | Field of study Electrical Engineering | Final grade 8.6/10 |

Thesis THE DEVELOPMENT OF A LOWER LIMB ROBOTIC PROSTHESIS FOR TRANSFEMORAL AMPUTEES

01/08/2023 – 26/06/2026 Vitória, Brazil

MASTER'S IN ELECTRICAL ENGINEERING Federal University of Espírito Santo

Website <https://www.ufes.br/> | Final grade 9.27/10 |

Thesis Control System for Robotic Lower-Limb Prostheses for Transfemoral Amputees Using Surface Electromyography of Antagonist Thigh Muscles

● LANGUAGE SKILLS

Mother tongue(s): PORTUGUESE

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● SKILLS

KiCAD | STM32 Cube IDE | MATLAB | C/C++ | Simulink | Basic Raspberry Pi | Python | Vicon | XSens | publish academic research | engineering principles | robotic components

● PUBLICATIONS

2025

[sEMG-Driven Robotic Prosthesis and Analysis of Muscle Activation in Transfemoral Amputees and Able-Bodied Individuals](#)

2024

Hardware development of a knee-ankle powered prosthesis for transfemoral amputees and its use in different ambulation modes.

P.H.F. Ulhoa, P.H.B. Buzatto, G.G. Fiorezi, A.B. Kock, T.F. Bastos-Filho and R.M. de Andrade. In 2024 XXIX Biomedical Engineering Brazilian Conference (CBEB).

Journal Name: IFMBE Proceedings

2024

[Design of a Powered Leg Prosthesis with a Custom Embedded Load Cell.](#)

Fiorezi, G. G., Ulhoa, P. H., Kock, A. B., Inacio, D. P., Buzatto, P. H., & de Andrade, R. M. (2024, September). 20th IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications. IEEE

2024

[EMG-based Co-Contraction Controller for an Upper-Limb Exoskeleton.](#)

Dias, E., Ulhoa, P., & Milanezi, R. (2024, September) 20th IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications. IEEE

2024

[Influence of Temporal and Frequency Selective Patterns Combined with CSP Layers on Performance in Exoskeleton-Assisted Motor Imagery Tasks](#)

Guerrero-Mendez, C. D., Blanco-Diaz, C. F., Rivera-Flor, H., Fabriz-Ulhoa, P. H., Fragoso-Dias, E. A., de Andrade, R. M., ... & Bastos-Filho, T. F. (2024). NeuroSci.

2023

[Experimental Evaluation of a Lightweight and Backdrivable Powered Knee Prosthesis for Transfemoral Amputees.](#)

Ulhoa, P.H., Fiorezi, G.G., Filho, T.B., Andrade, R.M. (2023). C3 IEEE Conference

2023

[Design of a 3 Degree-of-Freedom Upper-Limb Active Exoskeleton with Cable-Driven Actuators for Neuromotor Rehabilitation.](#)

Dias, E. A., Ulhoa, P.H., Andrade, R.M. (2023). C3 IEEE Conference.

2023

[Biomimetic Design of a Tendon-Driven Myoelectric Soft Hand Exoskeleton for Upper-Limb Rehabilitation.](#)

Silva, R. C., Lourenço, B. G., Ulhoa, P. H., Dias, E. A., da Cunha, F. L., Tonetto, C. P., ... & Andrade, R. M. (2023). Biomimetics, 8(3), 317.

2023

[Design and testing a highly backdrivable and kinematic compatible magneto-rheological knee exoskeleton.](#)

de Andrade, R. M., Fabriz Ulhoa, P. H., Fragoso Dias, E. A., Filho, A. B., & Vimieiro, C. B. S. (2023). Journal of Intelligent Material Systems and Structures, 34(6), 653-663.

2022

Designing a highly backdrivable and kinematic compatible magneto-rheological knee exoskeleton.

Andrade, R. M., Ulhoa, P. H., & Vimieiro, C. B. (2022, May). In 2022 International Conference on Robotics and Automation (ICRA) (pp. 5724-5730). IEEE.

2021

Digital Prototyping of a Microprocessor Controlled Active Knee Prosthesis

G. G. Fiorezi, P. H. F. Ulhoa, A. B. Filho and R. M. De Andrade, 2021 IEEE URUCON, Montevideo, Uruguay, 2021, pp. 517-521.

2020

Biomimetic design of a planar torsional spring to an active knee prosthesis actuator using FEM analysis.

Fiorezi, G. G., Moraes, J. D. S. D., Ulhoa, P. H. F., & Andrade, R. M. D. (2020, November). In Proceedings (Vol. 64, No. 1, p. 30). MDPI.

● **HONOURS AND AWARDS**

2023

Honors Mention by the Municipal Council of Vitória (Brazil) – Municipal Council of Vitória (Brazil)

Honors Mention by the Municipal Council of Vitória (Brazil) for the work in the development of a robotic lower limb prosthesis for transfemoral amputees. (2023).

1st Place - Innovation Competition

1st place at the Innovation Competition from the DNC group, competing with research projects from all over Brazil. The winning project was the Active Knee Prosthesis developed during my undergraduate studies.

2024

Honors Mention by Brazilian Society of Biomedical Engineering – Brazilian Society of Biomedical Engineering

Honors Mention in the XXIX Brazilian Conference of Biomedical Engineering (CBEB 2024) for the paper entitled "Hardware Development of a Knee-Ankle Robotic Prosthesis for Transfemoral Amputees and its Use in Different Ambulation Modes" based on the developments during my bachelor's project.